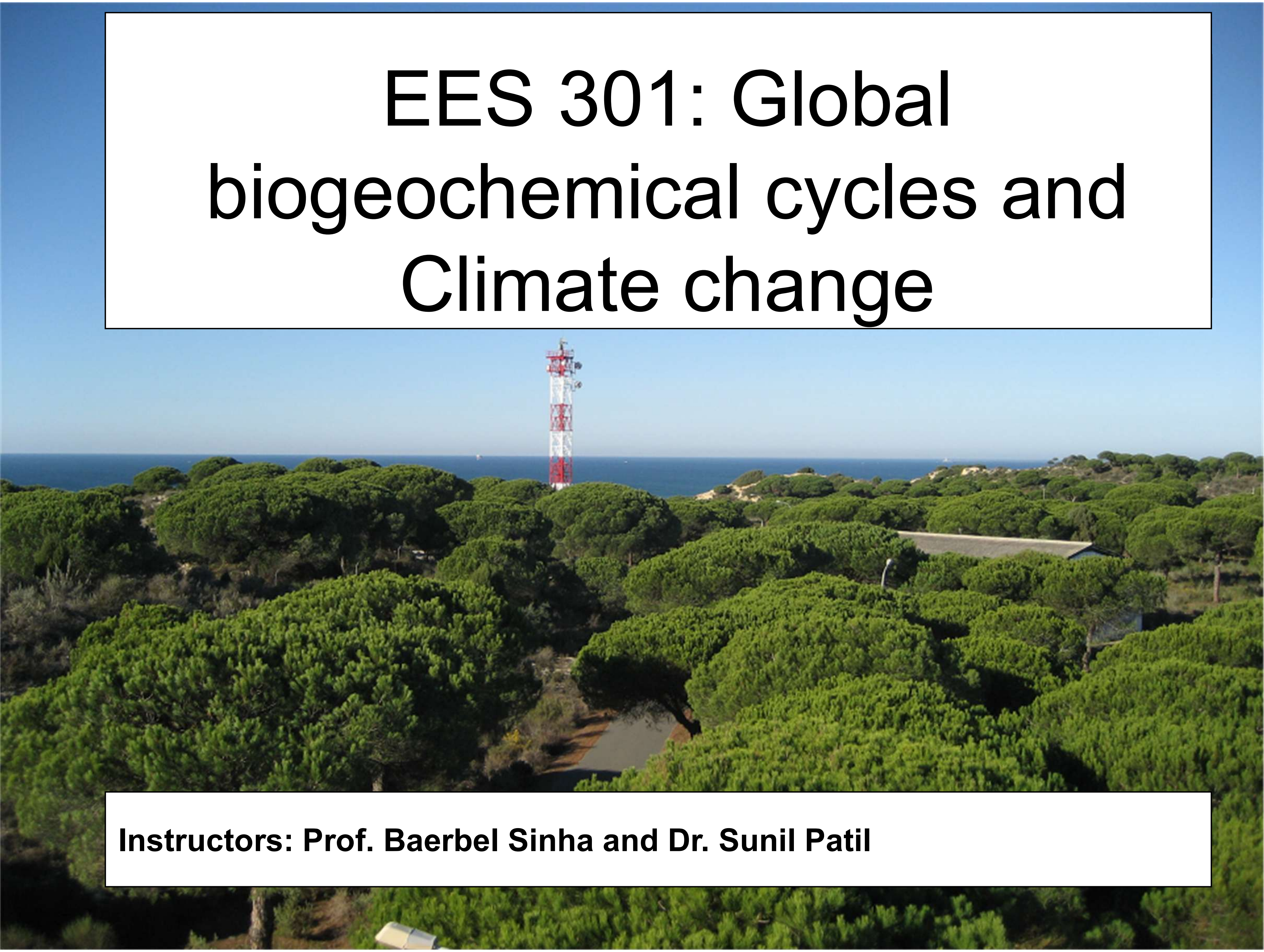
EES 301: Global biogeochemical cycles and Climate change



Instructors: Prof. Baerbel Sinha and Dr. Sunil Patil

IMPORTANT FACTS & Grading Policy

- Lecture slides: Will be uploaded on Moodle
- Textbooks: Have been ordered but may arrive late

- Exam practice & Problem solving: There are no solved problems in the textbook, and numericals/derivations may be done on the board rather than in the lecture slides. If you don't come to class or fail to take notes, you may find the exam very hard
- Tutorials (Friday class) will happen only after we have covered some syllabus and only on request from the class

Total Marks	:	100
Mid Sem Exam I	:	25 (Dr. Baerbel Sinha)
Mid Sem Exam II	:	25 (Dr. Baerbel Sinha)
End Sem Exam	:	50 (Dr. Sunil A. Patil)

Grading Relative

Recommended books

Recommended Books:

- **Michael Jacobson, Robert J Charlson, Henning Rodhe, Gordeon H Orians, Earth System Science: From Biogeochemical Cycles to Global Changes: Volume 72 (International Geophysics), ISBN 978-0123793706**
- **Schlesinger & Bernhardt, Biogeochemistry and analysis of global change, 3rd edition, Academic Press, ISBN 978-0123858740**
- **Dennis Hartmann, Global Physical Climatology, 2nd Edition, Elsevier Science**

Additional reading:

- **Berner & Berner, Global Environment, Water, Air, and geochemical cycles, Princeton**
- **University press, 2nd edition ISBN 978-0691136783**
- **Tim Lenton, Earth system science a very short introduction**

Syllabus - Before 1st Midsem

- **Introduction: The scientific method (4th August)**
- **Introduction: Earth System Science and Global Change (5th August)**
- **The origin and Evolution of Earth (7th & 11th August)**
- **Evolution and the Biosphere (12th & 14th August)**
- **Modelling Biogeochemical cycles (19th & 21st August)**
- **Equilibrium rate and Natural Systems (22nd & 25th August)**
- **Water and the Hydrosphere (26th & 28th August)**

Due to unavoidable circumstances the class of Monday 18th August needs to be shifted to the tutorial hour on Friday 22nd August

Syllabus between 1st Midsem & Midsem break

- The Atmosphere (4th & 8th September)
- The Ocean (9th & 11th September)
- The Climate System: Forcings, Feedbacks and responses (15th & 16th September)
- Ice Sheets and Ice Core record of Climate change (17th September)
- The Acid-Base and Oxidation Reduction balances of the Earth and their link to pollution (18th September)
- Human modifications of the Earth system: Acid rain, stratospheric ozone depletion & Global Change (22nd and 23rd September)
- Second midsem exam will be conducted in the regular class on 25th September (before midsem break)

Syllabus After Midsem break

- Soils, Watershed Processes and Marine Sediments
- Lithosphere and Biosphere
- The Global Carbon Cycle
- The Nitrogen Cycle
- The Sulfur Cycle
- The Phosphorous Cycle
- The Mercury Cycle
- Trace metal cycling (e.g., Fe, Mn)
- The Coupling of Biogeochemical cycles
- Human modifications of the Earth system: Large-scale eutrophication, food production

THINKING ABOUT ENVIRONMENTAL SCIENCE: WHAT DOES IT MEAN ?

The Planet Earth

- Unique in the universe (?)
- Mild, relatively constant temperatures
- Biogeochemical cycles
- Millions of species
- Diverse, self-sustaining communities
- A series of so-called Goldilocks conditions had to be met to create what we have
- Complex interlinked system where the whole is more than the sum of its parts

Scientific principles

The universe is sensible and governed by immutable rules.

Our job as scientists is to discover & understand these rules

Earth System Science is a science...

...just like chemistry and physics!

Geologists and Environmental scientists face the special

challenge of not being able to do

experiments in the sense that

chemists and physicists do.

The problem of experiments

Since geologists and environmental scientists are interested in systems that are very big (hundreds of km) and that have evolved over long periods of time (millions of years), they cannot conduct controlled experiments on the full system.

They must observe the results of Nature's/Mankinds experiments that are already complete.

Specific reaction or parts of the system can be studied with experiments.

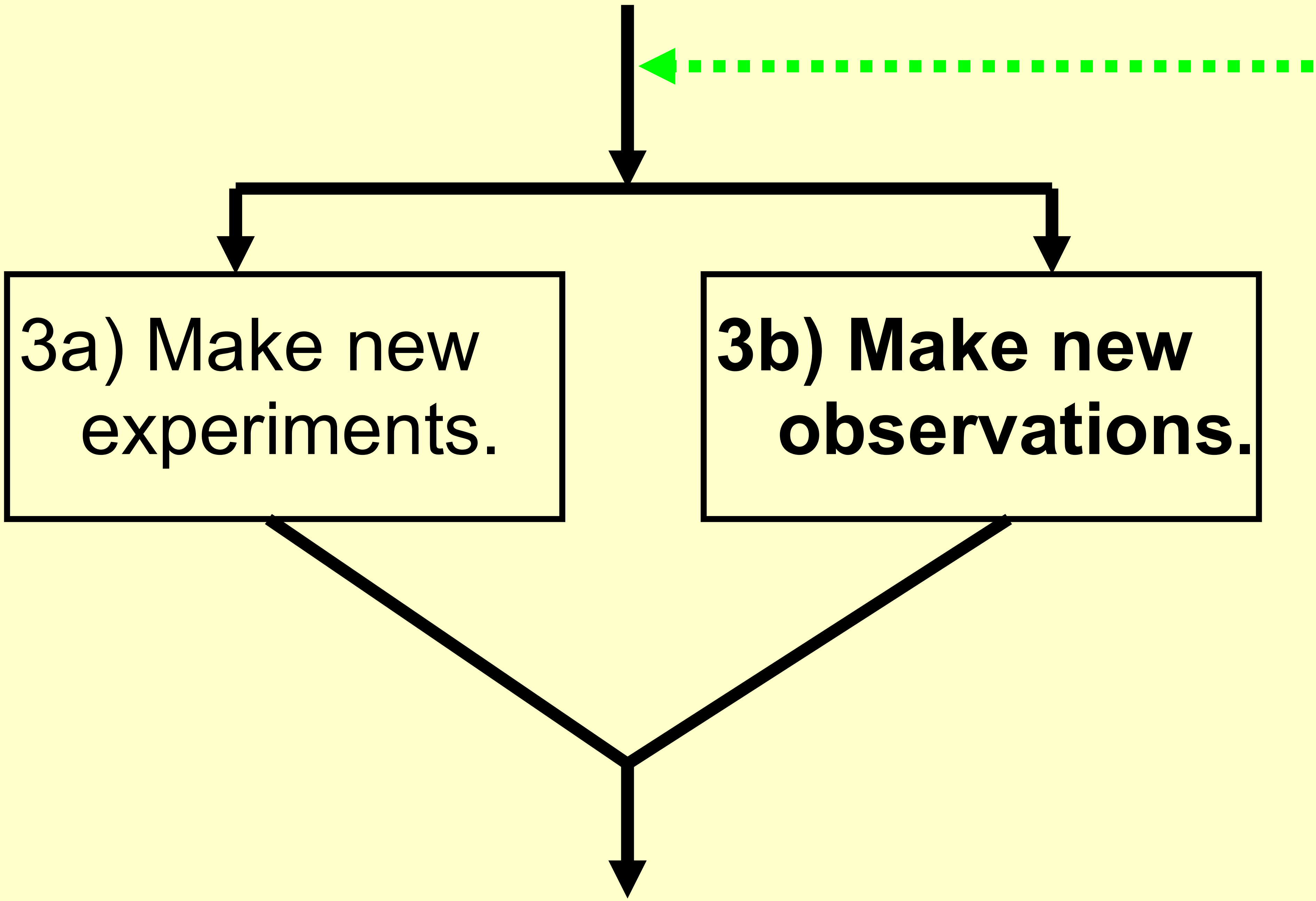
The inductive scientific method

1) Ask a question after making an observation about the sensible world.

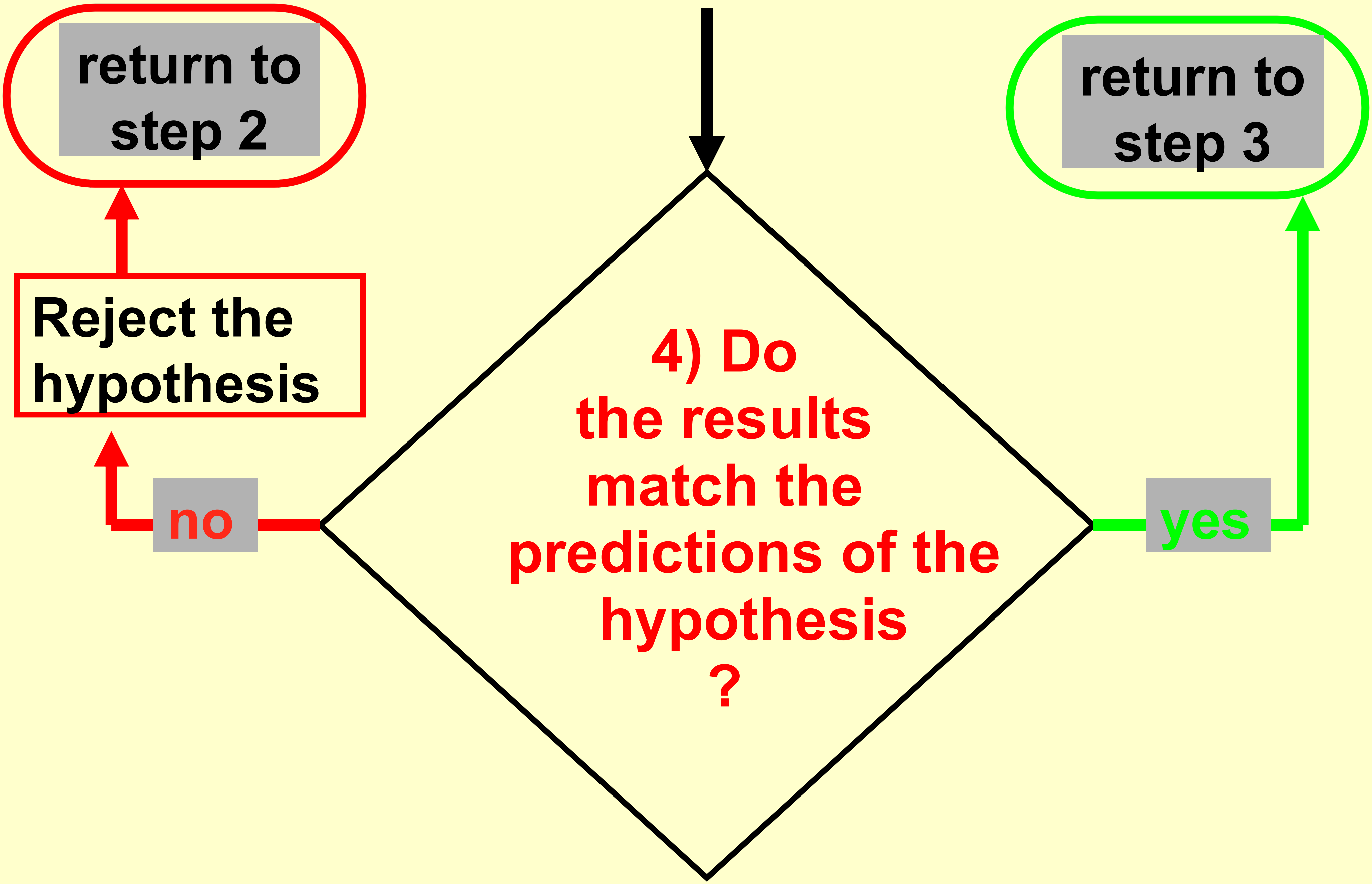
2) Develop an explanation. And educated first guess without sufficient evidence to be backed up by observations is called a conjecture. When we develop a model *that predicts* the outcome of other observations or experiments it is called a hypothesis.

Only with a model/hypothesis can we proceed to step 3

THE INDUCTIVE SCIENTIFIC METHOD (CONT.)



THE INDUCTIVE SCIENTIFIC METHOD (CONT.)



Hypothesis - Theory - Law

A hypothesis is an explanation initially offered for a set of observations. To be a hypothesis it must predict observation and needs to be testable.

When a hypothesis withstands many tests it may be called a theory.

A theory for which it seems there to be no sensible reasons to challenge is called a law.

A good theory is...

- *Parsimonious*
(is the simplest explanation available)

- *Consilient*
(explains a wide range of phenomena)

This scientific methodology is called induction

- It is inherent to the INDUCTIVE scientific methodology that theories can never be proven to be true! They can spectacularly fail to explain observed phenomena after years of confirmation => Newton's laws of mechanics at the beginning of the 20^{th} century
- Any proposal of about the universe that cannot be evaluated by observing the natural world is not a scientific theory or hypothesis it is a believe system.

When we conduct experiments we manipulate the independent variable and observe the dependent variable

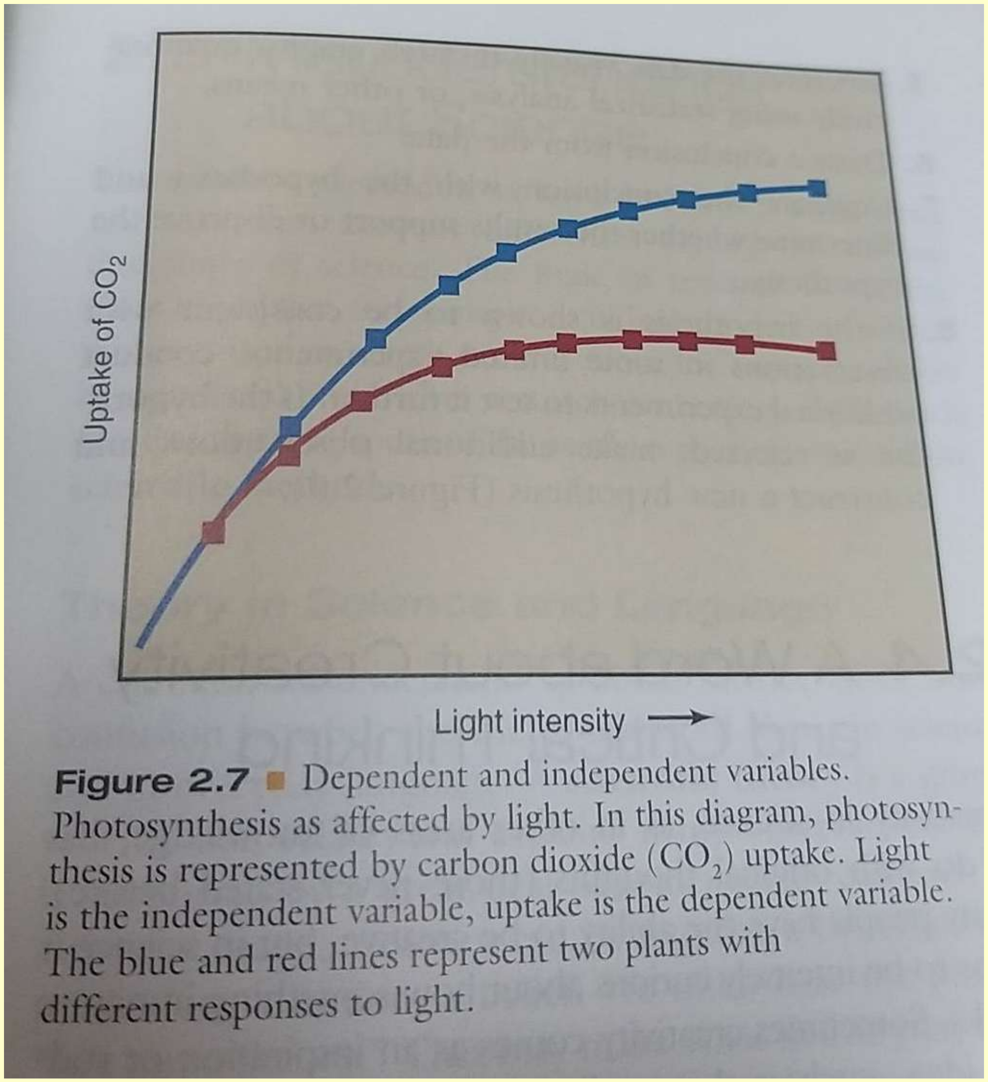


Figure 2.7 ■ Dependent and independent variables. Photosynthesis as affected by light. In this diagram, photosynthesis is represented by carbon dioxide (CO_2) uptake. Light is the independent variable, uptake is the dependent variable. The blue and red lines represent two plants with different responses to light.

Suppose someone wants the test the hypothesis that photosynthesis depends on light intensity.
They may place a plant in a plant chamber where the light level can be controlled accurately
They may observe how much carbon dioxide the plant takes up when exposed to a certain light level

In that experiment light intensity is the independent variable and CO_2 uptake the dependent variable

How your "hypothesis" ends up as H_A instead of H_0

- **Hypothesis testing - H_0 and H_A**
- **Suppose you want to test your hypothesis that light impacts plant growth: What do you do?**
- **Sadly mathematically you can only test "No difference between intervention and control" or "No effect of light"**
- **What you test becomes your H_0 hence:**
  your null hypothesis H_0 will be "Light has no impact on plant growth"
  What you believe to be the true hypothesis becomes your alternative hypothesis H_A – Light impacts plant growth
- **Now you set up an experiment keeping a control plant with sufficient light and an intervention plant in the dark/in low light and measure the difference in growth between the two plants. If your results indicate that H_0 "Light has no impact on plant growth" can be rejected with a 2σ confidence level, it is possible that your H_A is correct. However, mathematically there is no way to prove H_A. It is inherent to the inductive method that nothing can be proven correct. We can only prove a hypothesis wrong.**

The inductive scientific method

Simply put:

Always ask yourself,
“How will I know if I’m wrong?”

The role of scientific models in Earth system sciences:

1. Scientists develop numerical models as representations of how natural processes operates based on the best current scientific knowledge
2. Models are used to test the consistency of the knowledge and to make predictions
3. Predictions are then evaluated against observations to further evaluate and improve models (e.g. weather forecast models)
4. Models are used to simulate phenomena that are too big to replicate in a laboratory or operate over periods of times that are too long for humans to observe

PRACTICE QUESTIONS

You can submit the answer to these questions by Monday 11th August if you want to get them corrected before the Midsem exam

MCQ style

1) **What do you need to do to test the hypothesis “plants need water to grow”?**
   a) Design an experiment that could potentially proof the hypothesis plants need water to grow” to be wrong
   b) Look up the credibility of the person or source proposing the hypothesis
   c) Frame a testable null hypothesis “water does not impact plant growth” and your alternate hypothesis “water improves plant growth”, and design a suitable experiment to evaluate the null hypothesis
   d) Design an experiment that could prove the hypothesis “plants need water to grow” to be true

2) **Which statement best describes a hypothesis?**
   a) A hypothesis is a possible, testable explanation for a scientific question.
   b) The facts collected from an experiment are written in the form of a hypothesis.
   c) A hypothesis is the correct answer to a scientific question.
   d) A hypothesis is the process of making careful observations.

3) **What makes a model of the universe “scientific”?**
   a) It is accepted by government institutions.
   b) It is developed by scientists.
   c) It is primarily based on evidence.
   d) It is proven by experiment rather than observation.

MCQ style

1) Potato is an important food crop. Scientists want to perform an experiment to see if potato yield decreases when it is exposed to ozone, an air pollutant. How should the control group in this experiment be treated?
    a) The control group should have plants growing at higher oxygen levels.
    b) The control group should have plants growing in different amounts of water.
    c) The control group should have plants growing at different ozone levels.
    d) The control group should have plants growing in ozone free air.
2) Alysha wants to test the effect of different materials on the melting rate of ice. She finds the initial mass of four pieces of ice. She then covers three pieces with either tinfoil, plastic wrap, or paper. The fourth piece is left uncovered. After 30 minutes, Alysha calculates the final mass of each piece of ice to determine which melted the fastest. What are the independent and dependent variables in Alysha's experiment?
    a) The dependent variable is the mass of the ice, and the independent variable is the time the ice takes to melt.
    b) The dependent variable is the mass of the ice, and the independent variable is the type of cover material.
    c) The dependent variable is the time the ice takes to melt, and the independent variable is the type of cover material.
    d) The dependent variable is the type of cover material, and the independent variable is the mass of the ice.

Long answer style questions

1. Describe the inductive scientific method
2. Which role do numerical models play in $arth System Science
3. Explain the difference between a hypothesis, a theory and a law
4. What are the key properties of a good theory
5. Design an experiment to test the hypothesis “plants need water to grow” which involves a control and intervention group. Set up an appropriate null hypothesis and alternate hypothesis to test your hypothesis and designate your control and intervention group.